LICENSING LIFE-SAVING DRUGS FOR DEVELOPING COUNTRIES: EVIDENCE FROM THE MEDICINES PATENT POOL

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Abstract—We study the effects of the Medicines Patent Pool (MPP)—an institution that pools patents across geographical markets—on the licensing and adoption of life-saving drugs in low- and middle-income countries. We show the presence of an immediate and large increase in licensing when a patent is included in the MPP. We also show evidence that the pool increases actual entry and volume of sales, but these impacts are much smaller than on licensing, which is due to the geographic bundling of licenses. The paper highlights the potential of pools in promoting diffusion of biomedical innovation in developing countries.

I. Introduction

IN 2008 the U.S. Food and Drug Administration approved the use of etravirine, an HIV antiretroviral drug designed for patients with established resistance to other drugs. Yet, by 2015, etravirine was available in only about 25% of the lower- and upper-middle-income countries in Central and Eastern Europe. This is not atypical; other important HIV antiretrovirals and drug cocktails have limited availability in these countries (Gokengin et al., 2018). Nor are such delays limited to HIV drugs. Cockburn et al. (2016) show that global drug diffusion is very slow, especially in countries with relatively low incomes.

Promoting rapid and affordable access to essential drugs is a key health policy objective. One proposed approach is to use patent pools, which are voluntary arrangements in which patentees authorize the pool to license specific patents, typically as a bundle, to third parties. Historically, patent pools have been widely used in conjunction with technology standards—for example, in the electronic and information technology sectors—where the focus is on licensing complementary innovations. Such pools are designed to facilitate commercialization of standard-compliant products and follow-on innovation by lowering the transaction costs of licensing and coordinating the use of complementary innovations (Merges, 2001). More recently, a different type of patent pool has been proposed in the biomedical field, which focuses on promoting wider geographic diffusion of specific innovations rather than aggregating technologically related innovations. Examples include vaccines for the SARS epidemic, neglected tropical diseases and diagnostic testing (Van Zimmeren et al., 2011), and the COVID-19 patent pool recently proposed by the WHO.

Advocates of geographic patent pools argue that they can be particularly beneficial for promoting the diffusion of essential pharmaceutical products in small, low- and middle-income countries, where market entry may be at best marginally profitable. A specialized, centralized licensing platform may reduce the transaction costs of writing, monitoring, and enforcing patent contracts between drug companies and generic manufacturers, and thereby facilitate product launches in poor countries. However, skeptics claim that these pools are essentially public relations exercises, and that no evidence exists of real impact on access to medicines (Kennedy, 2015). They also argue that pharmaceutical companies can unilaterally facilitate diffusion in low-income countries by not filing, or committing not to enforce, patents in those markets. Other critics point out that the Trade-Related Aspects of Intellectual Property Rights (TRIPS) agreement allows countries to facilitate access by generic manufacturers through flexible provisions, such as compulsory licensing. These different views in the policy debate highlight the need for empirical research.

Evidence is limited on whether patent pools do, in practice, promote licensing and product launches because data on licensing contracts are typically confidential and difficult to obtain. In this paper, we exploit a rich data set on licensing from the Medicines Patent Pool (MPP) that allows us to conduct such empirical analysis. Established by the United Nations in 2010, the MPP is a voluntary licensing institution with a mandate to promote access to affordable and quality-assured treatments for HIV, tuberculosis, and hepatitis C in low- and middle-income countries. The MPP negotiates inclusion of patents in the pool with patent-holding pharmaceutical companies and then licenses the patent rights to interested generic drug companies. MPP licenses are nonexclusive, with low or zero royalty rates, and wide geographical scope.

We study how the MPP affects the licensing, launch, and sale of drugs in low- and middle-income countries. We begin with a short theoretical discussion of the differences between bilateral licensing by the patentee and a pool with geographical bundling. With a specialized patent pool as intermediary, the licensee is likely to face lower transaction costs but, at the same time, is constrained to license in all the countries designated by the pool. Such mandatory bundling reduces the correlation between licensing and drug launch because

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the pool is likely to include geographic markets where the licensee has no interest in launching. As we show later, this means that empirical studies of patent pools that focus only on licensing outcomes are likely to overestimate the effect on actual new product launches and welfare.

Our empirical analysis is based on panel data covering 173 pharmaceutical products and 129 countries over the period 2005–2018. The product set encompasses medicines for HIV, tuberculosis, and hepatitis C recognized as essential by the WHO. The sample covers countries classified by the World Bank as low- or middle-income and for which patent protection was in place for at least one of the sample drugs. We obtain licensing information for each product-country from the MPP. Crucially, the licensing information includes both MPP licenses and non-MPP bilateral licenses between the upstream patentee and generic firms, and covers both the medicines in the pool and other products in our sample. Moreover, licensing data are available from 2005, five years before the formation of the pool.

Our baseline empirical specification uses difference-in-differences and examines changes in patent licensing when product-country combinations are included in the MPP relative to those not included. The main endogeneity concern relates to the selection of products that enter into the MPP through upstream negotiations between the patent holder and the MPP. There may be unobservable factors that both are correlated with the inclusion of a medicine in the MPP and affect diffusion of the product across countries. For example, the MPP may be interested in the most effective drugs, which, in turn, would also be more attractive to generic licensees. To address this selection problem, we exploit the list of medicines that the MPP aimed to license when the pool was formed in 2010. This so-called "priority list" includes about 80 products. Fewer than half of these medicines eventually made it into the pool, but the MPP started a negotiation with the relevant patentees for all of them. This information allows us to estimate the effect of being included in the MPP by using as the control group those drugs for which bargaining with the pool started but failed. This identification strategy is closely related to the approach in Greenstone et al. (2010), who study the effect of large plant openings in a county using as a counterfactual counties that were considered by the plant managers but not chosen. We show that the estimates of the impact of the MPP using this approach are quantitatively similar to our baseline estimates.

We have three main empirical findings. First, the likelihood of observing at least one patent licensing deal covering a product-country combination increases substantially—by more than fivefold—once the associated patent is included in the MPP. This finding is robust to a wide variety of alternative specifications and controls. Second, the effect is heterogeneous: the increase in licensing due to the MPP is smaller in middle-income countries with high HIV incidence. These markets are more attractive for pharmaceutical companies, and bilateral deals are more likely to have been in place before the MPP. Finally, using additional data on product launches and sales for a subset of countries, we show that being included in the MPP also affects product market outcomes, not just licensing. The MPP is associated with higher entry rates (launches) of licensees, and this translates to higher quantities of products and lower prices.

The estimated impact of the MPP on market outcomes is much smaller than its effect on licensing. This is because of the geographic bundling of the MPP license contracts, as highlighted in our theoretical framework. This point is noteworthy because most studies of pools focus on how they affect licensing. But when licensing is bundled, this will overstate the impact on what ultimately matters, which is commercialization and product entry. Our estimates imply that, in our sample (which does not include sub-Saharan Africa), the probability of observing a licensing deal for product-countries in the MPP is more than five times larger than those not in the MPP, but the probability of observing an actual launch by a licensee increases by a more modest, but still substantial, 40%.

The paper is organized as follows. Section II discusses the related literature, and section III provides institutional details on the MPP. Section IV presents a simple conceptual framework. In section V we describe the data and the empirical specification. Section VI presents the baseline estimates of the effect of the MPP on patent licensing. Section VII examines the heterogeneity of the effect across countries. In section VIII, we investigate the impact of the MPP on drug launches and sales. We conclude with a brief summary of the findings and their implications.

II. Related Literature

Our paper is related to three strands in the literature on innovation. The first is the empirical literature on the diffusion of new drugs in developing countries. Cockburn et al. (2016) show that income levels, and patent and price regulation regimes, strongly affect how quickly new drugs are launched in a wide range of countries. Kyle (2006, 2007) examines the role of firm characteristics and price controls on pharmaceutical product launches in European countries. Delgado et al. (2013), Duggan et al. (2016), and Dubois et al. (2021) study how patent protection and centralized procurement affect drug sales and prices. Our paper complements this line of research by examining how a geographic patent pool can speed up the diffusion of essential drugs in low- and middle-income countries.

The second strand is the empirical literature on bargaining frictions in the market for technology (Furman & Stern, 2011; Williams, 2013; Galasso & Schankerman, 2015; Sampat & Williams, 2019). Furman and Stern (2011) show that the establishment of a public clearinghouse for biological resources (the U.S. Biological Research Center) promotes cumulative innovation. Williams (2013) shows that (contract-based) intellectual property protection on genes retards cumulative innovation, and Galasso and Schankerman

(2015) find that patent rights slow down cumulative innovation, albeit with very heterogeneous effects across technology areas. Our paper complements these studies by showing how a patent pool can reduce transaction costs and thereby promote licensing and technology diffusion. Our empirical approach is different, however, in that we rely on rich patent licensing data (for both patents included in the pool and related patents not included in the pool), as well as information on failed negotiations for inclusion in the pool, to identify causal effects.

Third, our paper relates to the literature on standards-based technology patent pools. The theoretical work studies incentives to form patent pools and the effect of such pools on innovation incentives (Lerner & Tirole, 2004; Quint, 2014; Reisinger & Tarantino, 2019). The empirical work is limited and focuses on two aspects: the factors that affect the participation decision, including the licensing and governance rules (Lerner et al., 2007; Layne & Lerner, 2011), and the impact of pools on innovation outcomes (Lampe & Moser, 2013; 2016).

Finally, two recent studies have looked at the MPP. Wang (2022) examines the impact of the MPP on pharmaceutical research expenditure, clinical trials, and sales of generic drugs. Martinelli et al. (2020) study the effect of MPP inclusion on the volume of drugs bought by procurement agencies and on the reallocation of market shares between originators and generics. Our paper complements these studies by focusing on the direct impact of patent pooling on licensing and its interaction with market outcomes. We also develop new identification strategies to address endogeneity of MPP patents and countries, which may also prove useful for future studies using MPP data.

III. The Medicines Patent Pool

The Medicines Patent Pool (MPP) is a voluntary licensing and patent pooling mechanism, established in 2010, with a mandate to accelerate access to affordable, appropriate, and quality-assured HIV treatments in developing countries. The global health organization UNITAID provided most of the financial support for the establishment of the MPP. In 2015 the mandate of the MPP was expanded to include hepatitis C and tuberculosis (Medicines Patent Pool, 2018).

The MPP negotiates patent licensing agreements with patent-holding pharmaceutical companies. The medicines to be licensed are identified in a priority list of drugs both approved and at the pipeline stage. The priority list is defined after consultations with the World Health Organization and experts from the government and the civil society. The choice of drugs takes into account medical needs and patent status in low- and middle-income countries (LMICs). Once a drug enters the MPP priority list, the MPP approaches relevant patent holders to negotiate for inclusion of the drug in the pool. In negotiating a license deal, the MPP focuses on the public health impact and aims to obtain broad geographical coverage and freedom to develop pediatric formulations and fixed-dose combinations that meet the needs of resource-limited settings. MPP licenses are typically royalty-free (more than 60% of licenses). In about 30% of the licenses royalties are paid to the patentee (typically 5% of net sales of finished products). In the remaining cases, royalties are collected by the MPP and channeled back to community-based HIV organizations in the country paying the royalty.[1]

Once a license deal with a patent holder is in place ("upstream license"), the MPP issues an expression of interest inviting generic producers to apply for sublicenses. In striking sublicenses, the goal of the MPP is to ensure that licensees have the capacity, willingness, and commitment to obtain appropriate regulatory approval, follow strict quality assurance requirements, develop the formulations, and make them available in the licensed territory. MPP sublicenses are nonexclusive and wide in geographical scope, including up to 131 low- and middle-income countries. Licenses often allow generic manufacturers to sell outside the licensed territory if they are not infringing on a patent.

IV. Theoretical Discussion

When the market for a pharmaceutical product is small—in terms of number of potential users and per capita income—the costs of negotiating, drafting, and enforcing a patent licensing contract may exceed the profits that can be extracted. In such cases, a drug company may still be interested in licensing its patents as an investment in corporate social responsibility (CSR) and virtue signaling, which generates indirect market value (Hong & Liskovich, 2015). A geographic patent pool can increase the private return to CSR investments by absorbing the costs of negotiating and monitoring licenses with generic manufacturers. Moreover, the information about the involvement of the patentee in poor countries may be disseminated more effectively by the pool.

When drug patents are included in a pool, downstream generic manufacturers negotiate with the pool for a bundled license that provides the right to sell the product in multiple countries. This can affect licensing outcomes and product launch decisions. In appendix A1, we develop a simple model illustrating the different implications of bilateral and pool licensing. The model is based on two key assumptions. First, we assume that pool licenses involve a fixed bundle of countries, whereas bilateral deals allow patentee and licensee to include in the contract the desired target countries. Second, we assume that the cost to the licensee of negotiating, drafting, and implementing a license contract might be lower when done through the pool as compared to a bilateral deal. The costs required for a bilateral license increase

[1]There is no within-firm variation in royalties in our sample. In particular, the 5% royalties involve all the HIV products licensed from Gilead, and the royalty channeled back locally is related to the only HIV drug licensed from Bristol Myers Squibb. This prevents us from identifying the effect of royalties as it would be collinear with the firm effect.

with the number of countries included, as the license become more complex with the involvement of patents in different jurisdictions and country-specific contractual provisions. Negotiating a license with the pool is less expensive than obtaining the full bundle of countries with a bilateral negotiation. In the context of the MPP, this difference is likely to be particularly large because most aspects of the MPP license contracts are standardized and do not require negotiations. In other contexts, where pool licenses are more tailored toward the specific licensees, the difference in transaction costs between bilateral and pool license may be much smaller.

We use this simple model to compare the conditions required to have licensing and commercialization across countries with and without a pool. There are two basic predictions. First, a geographic patent pool has an ambiguous effect on the total amount of licensing relative to bilateral contracting. When the profits from commercialization in each of the countries licensed in the pool bundle are large relative to the transaction costs, creating a patent pool has no effect on licensing as all patents would be licensed with or without a pool. When the profits that can be obtained across countries are asymmetric, the presence of a pool may increase or decrease the number of patents licensed. This is because with a bilateral deal the licensing firm chooses to license only patents associated with countries for which commercialization profits exceed licensing transaction costs. With the pool the licensee has to license the full bundle of countries, and this is profitable only if the corresponding transaction costs are not too large.

To see how this can lead to less licensing, consider a pool bundling N countries and a firm able to extract positive commercialization profits in only one of those countries. The bilateral deal would focus on only the profitable country. Assume that the commercialization profits made in this country are marginal and barely exceed the transaction costs required to strike the bilateral deal. In this case there may be no license if the firm is constrained to obtain the full bundle of countries from the pool, as the small commercialization profits (the same as in the bilateral deal) may be lower than the transaction costs associated with a contract covering the N patents for the N jurisdictions. Conversely, a pool may increase the total number of patents licensed when the transaction costs required to access the bundle of countries are small and close to (or lower than) those for the one-country bilateral license. This is likely to be the case in the context of the MPP, because most aspects of the bundled contracts are standardized and do not require negotiations.

The second implication of our model is that, in a patent pool with bundled licenses, the correlation between patent licensing and drug launch may be low. With costly bilateral licensing, the generic firm is expected to launch the product in any country for which it has a license. With a pool, the firm is constrained to license a bundle of countries, in some of which a launch may not be profitable. This observation highlights an important distinction between patent pools that license complementary technological components and pools like the MPP that bundle different geographical markets for a given technology. In the first case, one would expect that most, if not all, the licensed patents would be used by the licensee to ensure compliance with the technology standard. This does not apply in pools which serve as clearinghouses for international market access. As a consequence, empirical studies that rely on licensing data alone may overestimate the effect of a pool on actual commercialization.[2]

Appendix A1 discusses the model in greater detail and shows that the empirical predictions are robust to a variety of generalizations. These include allowing for the coexistence of MPP and bilateral licensing, differences in royalties between MPP and bilateral deals, and the presence of multiple licensees.

V. Data and Econometric Specification

A. *Data Construction*

The analysis starts with a balanced, product-country-year panel data set covering the period 2005–2018. The products in our sample were retrieved from three different sources. First, we download the products listed in MedsPal, a data portal managed by the MPP providing information on key drugs for HIV, hepatitis C, tuberculosis, and a few other diseases prominent in low- and middle-income countries. Second, we added to this list all the medicines in the 2017 WHO Essential Medicine List for HIV, tuberculosis, and hepatitis C that were not included in MedsPal. This list is a guide for national institutions in identifying products that need to be available in a functioning health system, and is based on disease prevalence, public health relevance, clinical efficacy, and cost effectiveness. Finally, we include products related to compounds in the 2010 MPP priority list, if these were not included in MedsPal or the Essential Medicine List.

The data set covers 216 products. Each product is defined by a molecule-strength combination (e.g., abacavir 300 mg or etravirine 100 mg). About half (52%) of the products are in the 2017 WHO Essential Medicine List. The 2010 MPP priority list includes 83 products. The countries in our sample are the 177 countries listed by MedsPal in July 2019, which includes all countries classified as low- or middle-income by the World Bank and some countries that, in recent years, graduated to high-income status. We collected information on the patents protecting products in each country. For 63% of the products in our sample, this information is provided by the MedsPal data set. For the remaining 37% (80 products), the information was retrieved manually from a variety of sources (including the DrugBank database,

[2]This also applies to studies in the public health literature (e.g., Juneja et al., 2017) that have tried to estimate the impact of the MPP assuming that its commercial impact takes place in all countries covered by its licenses. In practice, even a technology patent pool may contain patents that are not, strictly speaking, essential for the implementation of a standard. To this extent, some of the patents may not be implemented by all licensees (Lemley & Simcoe, 2018).

the WIPO Pat-INFORMED data, and Google Patents). Using this information, we identify product-country-years for which there is at least one nonexpired patent protecting the product in the country in the focal year.

Our main source of data for patent license agreements is the MPP. In this paper, for clarity, we refer to licensing deals between the MPP and pharmaceutical firms as "upstream licenses," and licensing deals between the MPP and downstream generic manufacturers as "sublicenses." Finally, we refer to the non-MPP deals between a pharmaceutical company and a generic producer as "bilateral licenses."

The MPP website provides access to the full text of their upstream licensing deals. In some cases, these agreements were renegotiated over time to include additional countries or drug formulations. The MPP shared with us all the historical contracts, allowing us to track how each upstream licensing deal evolved over time. The pool's website also provides information on sublicense agreements between the pool and generic manufacturers for all the drugs in the pool. The MPP also discloses, through its MedsPal database, information on bilateral licensing deals related to non-MPP drugs. This information is retrieved both from public sources (such as licensors' websites or official press releases) and from nonconfidential direct communications to the MPP from the licensors. The information on bilateral licences deals available on the MedsPal portal covers only deals signed after 2010, but the MPP also collected information on bilateral deals signed between 2001 and 2010 and shared it with us for this project.

The final data set is an unbalanced panel with 80,103 observations encompassing 129 countries and 173 products.[3] Focusing on product-country-years observations in which patent protection is in place allows us to examine the effect of inclusion of a patent in the MPP on the likelihood that the patent is licensed in the country. If anything, restricting to this sample leads to an underestimation of the effect of the pool if licensees also sell in countries where patent protection is not in place.[4]

B. *Descriptive Statistics*

Our empirical analysis focuses on two outcome variables. The first is *Deals*, which is defined as the total number of licensing deals (either bilateral or MPP sublicense) that are in force in the product-country-year observation. The second is a dummy variable, *Access*, which is equal to 1 if at least one of these licensing deals is in force in the product-country-year observation.[5]

TABLE 1.—DESCRIPTIVE STATISTICS

Panel A: Summary Statistics

	Obs.	Mean	SD	Min	Max
Access	80,103	0.180	0.385	0	1
Deals	14,453	4.589	4.365	1	20
MPP sublicenses	14,453	1.408	2.801	0	17
Bilateral deals	14,453	3.181	3.609	0	13

Panel B: Mean comparisons for MPP inclusion

	Obs.	Access	Deals
Product-countries never in MPP	65,886	0.133 (0.339)	0.481 (1.842)
MPP product-countries before entry	8,588	0.092 (0.290)	0.454 (1.857)
MPP product-countries after entry	5,629	0.870 (0.335)	5.454 (5.021)

The unit of observation is product-country-year. In panel A the statistics for Deals, MPP sublicenses, and bilateral deals are restricted to the sample in which Access = 1. In panel B standard deviations are in parentheses.

Panel A of table 1 provides summary statistics for some of the key variables in our data set. The dummy *Access* equals one for about 18% of the product-country-year observations in our sample. Conditional on having at least one license in place, the average number of *Deals* is 4.6. Of these deals, 1.4 are MPP sublicenses and 3.2 are bilateral deals. By the end of 2018, about 27% (47/173) of the products in our sample are included in the MPP.

Appendix A2 provides additional descriptive analysis comparing the geographical scope of MPP licenses with non-MPP bilateral agreements that are related to the same set of products. Summarizing briefly here, the average number of countries covered by MPP licenses is 24 (median is nine), whereas the mean for a bilateral license is 15 (median is two). This ranking holds more generally—the distribution of geographical scope in the MPP licenses stochastically dominates the one for bilateral deals. Moreover, this comparison actually underestimates the extra geographic coverage—what we call "geographic additionality"—provided by MPP licenses.[6] Examining the identity of the countries covered,

[3]We drop products that are listed in MedsPal with patents expired before 2005, those with patents filed after 2018, and products that are not in MedsPal and for which patent information could not be retrieved from other data sets. We drop countries for which our data show no patent filed for all of the products during the entire sample period. These include conflict-affected poor countries (such as Afghanistan and South Sudan) and a number of very small countries (such as the Cook Islands, Tonga, and Micronesia).

[4]Typically, MPP license deals allow licensees to sell outside the licensed territory as long as they are not infringing on any granted patents. Our discussions with MPP executives suggest that this provision is not present in many bilateral (non-MPP) deals that may include stricter restrictions on sales.

[5]Although the number of MPP sublicenses is accurately reported, for some of the bilateral deals we do not know the exact number of licensees involved (but we do know the products involved and the geographic coverage of the deal). This required us to perform two types of imputations in our data. First, we set the number of deals as equal to one in cases where no information on the number of licensees is reported in our data. Second, in some of these cases licensing deals are expanded geographically, and we assume that that the revised agreement involves all the original licensees. In appendix A3, we show that adding a dummy for these adjustments or dropping these subsamples has no impact on our findings.

[6]To illustrate, consider a product for which the MPP license covers 24 countries, and the bilateral license covers fifteen countries. The difference in the number of countries covered captures the additionality only if all the countries included in the bilateral agreement are also included in the MPP license. If the MPP license includes only a subset of those covered by the bilateral deal, a simple count of countries will underestimate the MPP additionality. In the extreme case where there is no overlap between the countries in the MPP and bilateral licenses, the MPP additionality is 24

we found that, on average, MPP licenses include 9.9 countries in excess of the bilateral deals covering the same product; the median number is three countries. Finally, we show that the MPP additionality is much larger in the lower-middle-income countries, as compared to low-income or upper-middle-income countries, and this is even more pronounced when countries are weighted by population.

Panel B of table 1 provides preliminary evidence of the effect of a country-product dyad entering the pool. The first row shows the baseline probability of at least one MPP license (*Access*) and the average number of deals for country-product combinations that are not included in the MPP during our entire sample period. The second row presents the same information for country-products that enter the pool, for the sample years before their inclusion in the MPP. The last row provides similar statistics for country-products in the pool during the sample period in which they are included in the MPP.

The results are striking. Access rates are, respectively, 0.13, 0.09, and 0.87, implying that the probability of striking at least one license deal related to the product-country is nearly ten times larger once the product-country is included in the MPP. Similarly, the average number of deals are 0.48, 0.45, and 5.45. These are very large effects, but they may confound the causal effect of pool inclusion with unobserved heterogeneity driven by country and product characteristics. The econometric analysis below attempts to control for these dimensions of heterogeneity. Moreover, as emphasized earlier, given the bundled licensing by the MPP, patent license deals capture "potential generic entrants" for the product-country rather than actual entry. In section VIII we examine this distinction in detail, using drug launches and sales data purchased from a private vendor.

C. Empirical Model and Identification

Our empirical strategy relies on a difference-in-differences specification:

$$Y_{p,c,t} = \alpha + \beta MPP_{p,c,t} + \gamma X_{c,t} + \delta_t + f_{p,c} + \varepsilon_{p,c,t}, \qquad (1)$$

where the dependent variable, $Y_{p,c,t}$, captures licensing activities in product, p, country, c, and year, t. The unit of observation is a product-country-year. The treatment variable, $MPP_{p,c,t}$ is equal to 1 for product-country dyads that are included in an upstream license between the MPP and a pharmaceutical firm in year t. The term $X_{c,t}$ captures a series of time-varying country controls. The terms δ_t and $f_{p,c}$ are year and product-country fixed effects. The coefficient β is the difference-in-differences estimator identifying the effect of entering the pool on licensing relative to product-countries that are not in the MPP. The baseline regressions are estimated by OLS, with standard errors clustered at the product and country level.

countries, which is much larger than the simple comparison of the number of countries covered, which is nine in this example.

One potential concern with our empirical approach is that products are not likely to be randomly allocated to the patent pool. There may be unobservable factors that are correlated with both the likelihood that the product enters the MPP and the underlying demand for licenses for that product. The direction of the induced bias is ambiguous, however, and would depend on how the MPP decides which products to target. For example, the MPP may decide not to target the most effective drugs with large demand, as it may anticipate that there would be licenses for these products even without their intervention. This would induce a negative bias in the OLS regression of licensing against MPP inclusion. At the same time, the MPP may not consider drugs with niche markets and small demand, as it might expect no licensing interest even if the drugs were included in the MPP. This would induce a positive bias in our estimates.

Our identification strategy to address this issue is to use the drugs for which negotiations with the MPP failed as a counterfactual for what would have happened to the drug that entered the pool in the absence of pool inclusion. To conduct such an analysis, we exploit the MPP 2010 priority list. This list of medicines was compiled when the pool was established, after consultations with the WHO, national governments, and other experts. It comprises medicines that the MPP wanted to be included in the pool in 2010. For all these drugs, MPP entered a negotiation with the pharmaceutical company owning the patent. The 2010 priority list encompasses 83 of our sample products, but only 38.5% of them eventually made it into the MPP. Under the assumption that success or failure in the negotiation for these drugs was quasi-random—that is, not related to unobservable drivers of future licensing—focusing on this priority list would alleviate concerns related to selection into the MPP.[7] The key identification assumption for this analysis is that drugs in the 2010 priority list would have trended identically in the absence of MPP inclusion, conditional on the other control variables. This identification strategy is similar to the one employed by Greenstone et al. (2010), who estimate the effect of large plant openings in a county using counties that were considered by the plant managers, but not chosen, as a counterfactual. It also resembles the approach in Seru (2014), who exploits failed mergers to generate exogenous variation in acquisition outcomes of target firms.

We complement the standard difference-in-differences estimation equation (1) with the more flexible dynamic specification:

$$Y_{p,c,t} = \alpha + \sum_t \beta_{t-\tau} MPP_{p,c,t-\tau} + \gamma X_{c,t} + \delta_t + f_{p,c} + \varepsilon_{p,c,t}, \qquad (2)$$

[7]Using failed negotiations over drugs in the priority list should also take care of another possible concern. Specifically, a licensee might decide to delay a bilateral negotiation over a product-country once it learns that the MPP is negotiating with a pharmaceutical company. This information is available at the same time for both treatment and control group, and so such strategic delay should not cause any bias.

TABLE 2.—MPP, ACCESS, AND LICENSING DEALS

Dep. var.	(1) Access	(2) Deals	(3) Access	(4) Deals	(5) Access	(6) Deals
MPP	0.663***	4.610***	0.603***	3.083***	0.588***	3.730***
	(0.054)	(0.779)	(0.062)	(0.615)	(0.052)	(0.526)
Sample	Full	Full	2010 priority list	2010 priority list	Full	Full
Year effects	Yes	Yes	Yes	Yes	Yes	Yes
Product-country effects	Yes	Yes	Yes	Yes	Yes	Yes
Product-year effects	No	No	No	No	Yes	Yes
Country-year effects	No	No	No	No	Yes	Yes
Observations	80,101	80,101	40,534	40,534	80,052	80,052

Standard errors clustered at the product and country level in parentheses. $^{*}p < 0.1$, $^{**}p < 0.05$, and $^{***}p < 0.01$. Deals = total number of licensing deals for the country product in year t (includes MPP and non-MPP deals). Access = 1 if Deals > 0. MPP = 1 if the product-country is included in an upstream MPP license. Columns 3 and 4 include only products in the 2010 priority list.

where τ is the year in which the product-country is included in the MPP. The dummies $MPP_{p,c,t-\tau} = 1$ at $t - \tau$ years from inclusion and 0 otherwise. Notice that $t - \tau$ includes leads and lags. In this extended model the coefficients $\beta_{t-\tau}$ capture the differences between the treatment and control product-countries for each year before and after inclusion in the MPP. Following convention, we take as the baseline effect the year before inclusion, (i.e., we normalize by setting $\beta_{-1} = 0$).

VI. Empirical Results

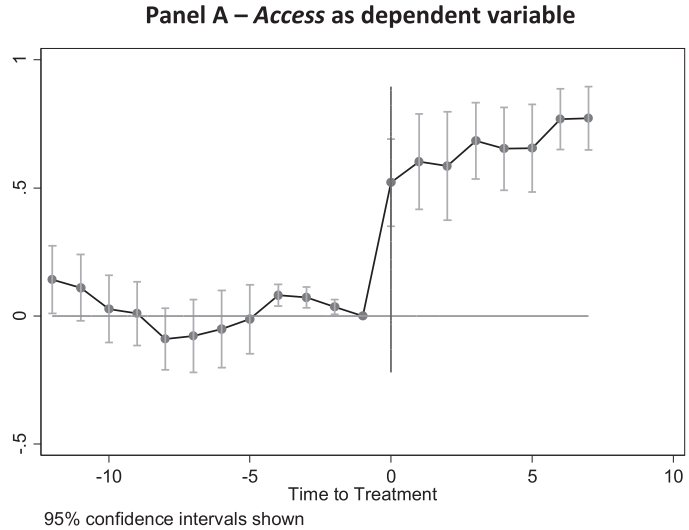
A. *Baseline Specification*

Table 2 provides the baseline results. In column 1, the dependent variable is *Access*, an indicator equal to 1 if at least one licensing deal is in place for product p and country c in year t. The estimated β is positive and statistically significant: the likelihood of at least one deal for the product-country increases by 66 percentage points after the product-country combination enters the MPP. The mean value of *Access* for non-MPP products-countries is 0.13, and so this corresponds to a fivefold increase in the likelihood of observing at least one licensing deal. In column 2 the dependent variable is the total number of licensing deals. The estimated MPP treatment effect is an increase of 4.6 deals, and it is strongly significant. Compared to the mean number of deals for product-countries that never enter the MPP, this represents a tenfold increase.

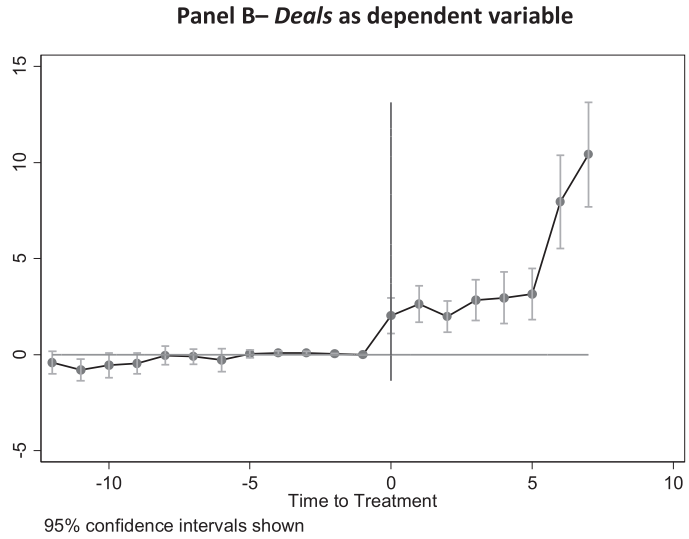
Columns 3 and 4 estimate the model on the sample of products on the MPP priority list of 2010. This approach uses the drugs for which negotiations failed as a counterfactual for what would have happened to the drug that entered the pool in the absence of pool inclusion. The MPP treatment effect on *Access* and the number of deals are similar to those in our baseline regressions, indicating that product-level selection into the MPP does not seem to be a source of bias.

Using the MPP priority list sample, we also estimate the dynamic specification described in equation (2). Panel A in figure 1 plots the estimated coefficients and their 95% confidence intervals, using *Access* as the dependent variable. The results confirm that inclusion in the MPP is associated

FIGURE 1.—DYNAMIC EFFECTS OF MPP INCLUSION



This figure plots the estimates of a regression in which the effect of the MPP is separately estimated for each year before and after inclusion. The figures plot the coefficients (and 95% confidence intervals) with the year before inclusion normalized to zero. Standard errors are clustered at the product and country level. The dependent variable *Access* is equal to 1 if the total number of licensing deals for the country product in year t is positive. The sample includes only products in the 2010 MPP priority list.



This figure plots the estimates of a regression in which the effect of the MPP is separately estimated for each year before and after inclusion. The figures plot the coefficients (and 95% confidence intervals) with the year before inclusion normalized to zero. Standard errors are clustered at the product and country level. The dependent variable is equal to the total number of licensing deals for the country product in year t. The sample includes only products in the 2010 MPP priority list.

with a large increase in the likelihood of licensing. The coefficients estimated for the period before inclusion are very small in magnitude and mostly statistically insignificant, broadly supporting the common-trends assumption required for identification in diff-in-diff models. There is clear evidence of a sharp, large, and immediate increase in patent licensing after MPP inclusion, and the estimated increase in licensing is statistically significant. This pattern confirms that MPP patented products are quickly involved in licensing deals after introduction in the pool.

Panel B in figure 1 plots the estimates for the specification using the number of *Deals* as the dependent variable. Here too we see a substantial and immediate increase in licensing deals after inclusion in the MPP, and the graph suggests that the effect is not driven by preexisting trends. Figure A2 in the appendix provides several robustness checks for this visual analysis of the dynamic effects of MPP inclusion.

As an alternative approach to address unobserved heterogeneity, in columns 5 and 6 we estimated a model including a full set of country-year and product-year effects in the full sample. These controls capture time-varying factors related to the economy of each country or the international market success of each product. In this more demanding specification, the size of the MPP coefficients falls by about 15%–20% relative to the estimates in columns 1 and 2, but the MPP effect remains statistically significant and economically large.

B. Robustness

We perform a variety of robustness checks on our main findings. These include further analysis of the priority list sample, using alternative econometric specifications and dependent variables, and conducting the estimation at the molecule, rather than the product, level of analysis. The details of these (and other) empirical tests are presented in the online appendix A3. In this subsection we briefly summarize two main robustness checks related to the staggered treatment rollout and the endogeneity of the countries included in the MPP licenses.

Decomposition of the treatment effect using the imputation estimator. Our empirical setting differs from the standard difference-in-differences setup because units in the panel are treated (included in the MPP) at different points in time. Recent studies in the econometrics literature have shown that identification problems can arise when treatment rollout is staggered (see Baker et al., 2021 for a survey). The resulting biases are more severe when the estimation relies heavily on a comparison of units that are treated over a period of time and "reference" units that are treated earlier in the sample, relative to completely untreated units in the control sample. In our setting, the size of the treatment group is small relative to the control group, so most of the comparison is likely to be between drugs included in the MPP and the control drugs, rather than between those included in the MPP earlier rather than later, so the bias may be limited.

We address this concern in two ways. First, we decompose our two-way fixed effects estimator, following the procedure developed by Goodman-Bacon et al. (2021). The decomposition confirms that the timing variation accounts for less than 4% of the estimated effect, and that essentially all the identifying variation comes from comparisons to product-country pairs that are not included in the MPP during our sample period. Second, we reestimate our baseline model using the "imputation estimator" developed by Borusyak et al. (forthcoming), which is robust to treatment effect heterogeneity in panels with staggered rollout. This estimator computes unit and period fixed effects using only untreated observations, and then generates imputed, untreated potential outcomes to estimate treatment effect for treated observations. As it turns out, the estimated coefficients (reported in appendix table A1) are essentially identical to our baseline coefficients.

Endogeneity of country coverage. Our baseline regression includes product-country fixed effects that account for unobserved, time-invariant characteristics that may be correlated with both upstream and downstream licensing. However, these could be time-varying country characteristics that affect both the inclusion of countries in MPP licenses and bilateral licensing. For example, government policies that make the market more attractive to a patent-holding drug company may make bilateral licensing more likely and thus make it less likely that the country will be covered by the upstream license with the MPP. To address this potential endogeneity of country coverage, we exploit the annual World Bank classification of countries by income level. When negotiating the geographical scope of a deal, the MPP focuses primarily on low-income and lower-middle-income countries, though occasionally upper-middle-income countries may be covered.[8]

In appendix A3, we first confirm that the likelihood of being included in the MPP is much higher for lower-middle-income, as compared to upper-middle-income, countries. Based on this, we exploit a fuzzy regression discontinuity design to estimate the treatment effect, using a dummy for whether the country is below the upper-middle-income threshold at the time of negotiations as the instrument for inclusion in the pool. The key identifying assumption requires that the World Bank classification, for countries near the threshold, does not affect licensing negotiations directly——that is, through channels other than the MPP—which seems reasonable in this context. Because this RDD analysis uses only observations in the neighborhood of the income

[8]This is both because drug companies see greater potential for bilateral deals for the upper-middle-income group (e.g., Brazil or China) and because the MPP itself has less interest in such countries (Branigan, 2018). We confirmed this directly with MPP executives, who reported that companies often use the upper-middle-income thresholds as a key criterion for agreeing whether to include a country into the upstream license.

TABLE 3.—HETEROGENEOUS EFFECTS

Dep. var.	(1) Access	(2) Access	(3) Access	(4) Access
MPP	0.646***	0.646***	0.591***	0.650***
	(0.058)	(0.058)	(0.064)	(0.053)
MPP × High disease prevalence	−0.056	0.004	−0.018	0.027
	(0.039)	(0.033)	(0.047)	(0.028)
MPP × High disease prevalence × Middle income		−0.139***	−0.176***	−0.103**
		(0.041)	(0.037)	(0.048)
Year effects	Yes	Yes	Yes	Yes
Product-country effects	Yes	Yes	Yes	Yes
Sample	HIV products	HIV products	2010 Priority List	All products
Observations	51,613	51,613	33,444	72,524

Robust standard errors clustered at the product and country level in parentheses. * $p < 0.1$, ** $p < 0.05$, and *** $p < 0.01$. Access = 1 if Deals > 0. MPP = 1 if the product-country is included in an upstream MPP license. High disease prevalence = countries in the top decile for number of people with disease related to the product in the year. Middle income =1 for countries classified by the World Bank as middle income. Models include direct effects for high disease burden and middle income.

threshold, it relies on a much smaller sample. The first stage regression confirms a strong negative correlation between the upper-middle-income status of the country and inclusion in the MPP. The estimated treatment effect of the MPP, using IV estimation, is slightly larger than the OLS estimate, but the difference is not statistically significant (see appendix A3 for details). This analysis suggests that endogenous country coverage does not seem to bias our results.

VII. Heterogeneous Treatment Effects

In this section we examine whether the impact of the MPP varies across countries. In the online appendix A4 we discuss additional evidence of heterogeneous effects across patentees and products.

From a theoretical perspective, we expect the MPP to have a smaller impact on licensing in geographical markets where patent owners have enough incentives to license their patents bilaterally. These incentives can come from two different sources: sufficient market size that makes royalties large enough to enter, or markets with characteristics that generate indirect market value through corporate social responsibility (CSR) and virtue signaling (Hong & Liskovich, 2015). To be at play, both the royalty channel and the CSR-signaling channel require a large population affected by the virus, as countries with limited demand for HIV medicines are unlikely to generate substantial revenue or halo effects. For a given level of HIV population, higher per capita income should strengthen the royalty motive, but weaken the CSR motive, for bilateral deals.

Therefore, to assess the strength on these two mechanisms, we compare the MPP treatment effect in low-income versus high-income countries where both have a large viraemic (HIV) population. If bilateral deals are driven by CSR motives, the treatment effect of the MPP on licensing should be *smaller in the low-income* countries, whereas it should be *smaller in middle-income* countries if bilateral deals are driven by direct financial returns.

We begin by examining whether the MPP treatment effect on *Access* varies across countries with different levels of HIV population. For this we include in this sample only products related to HIV. Using the full sample, column 1 in table 3 distinguishes countries in the top 10% in terms of population living with HIV (retrieved from the World Bank data portal) versus the rest. The estimated MPP treatment effect is about 5.6 percentage points smaller for the countries with the largest viraemic population, but the coefficient is not statistically significant.[9]

To distinguish between the royalty and CSR motives, column 2 splits the countries in the top decile of population with HIV into two groups: low-income and middle-income countries, using the World Bank classification. The results indicate that the impact of the MPP is 21% smaller in countries where there is both a sizable viraemic population (potential market) *and* sufficient ability to pay (effective market). In these cases, the market is profitable enough to induce bilateral deals and thus reduces the impact of the MPP on the probability that at least one license is observed. In column 3, we restrict the sample to the priority list drugs, and the results are even stronger—the MPP impact is now 30% smaller in markets with large HIV population, middle-income countries. Interestingly, in all these regressions the estimated coefficient on the interaction of the MPP dummy with a large HIV population is essentially zero and highly insignificant. This suggests that halo effects are not large enough to incentivize bilateral deals in the poorest countries.

Finally, in column 4 we enlarge the sample to include products related to tuberculosis (TB) and hepatitis C (HepC). We construct a measure of each country's population that is affected by TB and HepC, and, as before, we separate countries for which the viraemic for the disease targeted by the product is above the 90th percentile of the sample.[10] Adding these additional products does not change the conclusion from our baseline with only HIV products. The impact of

[9]We confirm this finding in a more flexible specification that includes dummies for various deciles of the distribution of viraemic population. We again find a negative interaction between MPP and the level of HIV population, with effects larger in the higher deciles, but these estimates are not statistically significant in most specifications.

[10]We use data on the incidence of TB per 100,000 people and country population provided by the World Bank. The World Bank data portal does not provide data on HepC. To construct a measure for HepC, we use the estimates in Blach et al. (2017) for the prevalence of viraemic HepC at the end of 2015 across 100 countries. Notice that these measures are not available for all our sample countries, so the sample size is smaller than our baseline analysis.

the MPP on *Access* is again lower in countries where disease prevalence and income level are large enough to stimulate bilateral licensing.

Our analysis focused on *Access* rather than the number of deals as the dependent variable. Coefficients on the interaction of MPP and country characteristics tend to be small and statistically insignificant when we use *Deals* as dependent variable. The main reason is that MPP downstream deals give licensees access to a bundle of countries. Therefore, any additional MPP licensee increases the number of deals in every MPP country covered by the license by one, independently of country characteristics. If there is only limited substitution with bilateral deals, the effect of the MPP on the total number of licensing deals will be more similar across heterogeneous countries.[11]

These results show that the MPP is more important in promoting greater access to poor countries and in middle-income countries with lower exposure to HIV. In this respect, the patent pool represents a gateway to needed drugs for countries that are not big enough to be an attractive market for bilateral deals.

VIII. Impact on Market Outcomes

Thus far we focused on the impact of the MPP on downstream licensing deals. However, these contracts give licensees the right to practice the patented product, but they may not necessarily translate into actual launch and sales. In short, patent license deals measure potential entry of the generic firms rather than actual entry. This distinction is important in our setting where MPP license contracts include large bundles of countries, as licensees are likely to only be interested in a subset of the countries.

To examine this issue requires data on actual entry by generic firms and their sales across the countries in our sample. We purchased a data set with this information from a private vendor, IQVIA, a leading provider of data on international sales of pharmaceutical products. The IQVIA data provide detailed information on the launch of new products across countries, sales revenues (in U.S. dollars), and volume (effective number of tablets) for the period of our study. Despite this level of detail, the IQVIA data do not cover our full sample. The sales data are available for only 32 countries out of the 129 countries in our sample, mostly middle-income countries outside Sub-Saharan Africa (such as Egypt, Pakistan, Peru, and Vietnam), and about 80% of the products in our sample.[12] Matching this information with our licensing data set, we obtain an unbalanced panel with 24,663 product-country-year observations.[13]

A. Impact on Entry, Volume, and Sales

We estimate how inclusion in the MPP of a country-product affects three market outcomes: product launch (entry), quantity sold, and sales revenue. To do this, we estimate the dynamic diff-in-diff specification given in equation (2).

Panel A in figure 2 plots the estimated coefficients and their 90% confidence intervals, where the dependent variable is a dummy equal to 1 if at least one licensee has launched the product in the country in the focal year. Panels B and C do the same, using the logarithm of sales revenue and volume of product sold, respectively.[14] We use the full sample of products in these regressions, but the results are very similar when we restrict the sample to those on the MPP priority list, as we report in appendix figure A4. Even with the full IQVIA sample of products, the limited number of launches observed (most products entered the MPP late in the sample period) makes it hard to estimate any impacts precisely. Moreover, we use conservative standard errors (clustered by product and country), which further contributes to large confidence intervals. Nonetheless, we think the results are a useful first step.

Panel A strongly suggests that inclusion in the MPP increases the likelihood that at least one licensee launches the product in the country, but this entry takes place with about a five-year lag. We find no evidence of any pre-MPP effects, consistent with the parallel trends assumption. The observed entry delay is consistent with information reported on the MPP website and publications on the time needed for licensees to sell once a patent deal is in place. Launch requires obtaining WHO prequalification of the generic version of the drug and setting up manufacturing and distribution facilities in the country.

Panel B shows that there is no statistically significant effect of being included in the MPP on *sales revenue*. There is also no evidence of any pre-trend effects. However, as panel C shows, evidence suggests that the MPP is associated with a steady increase in the *quantity sold* after a short lag,

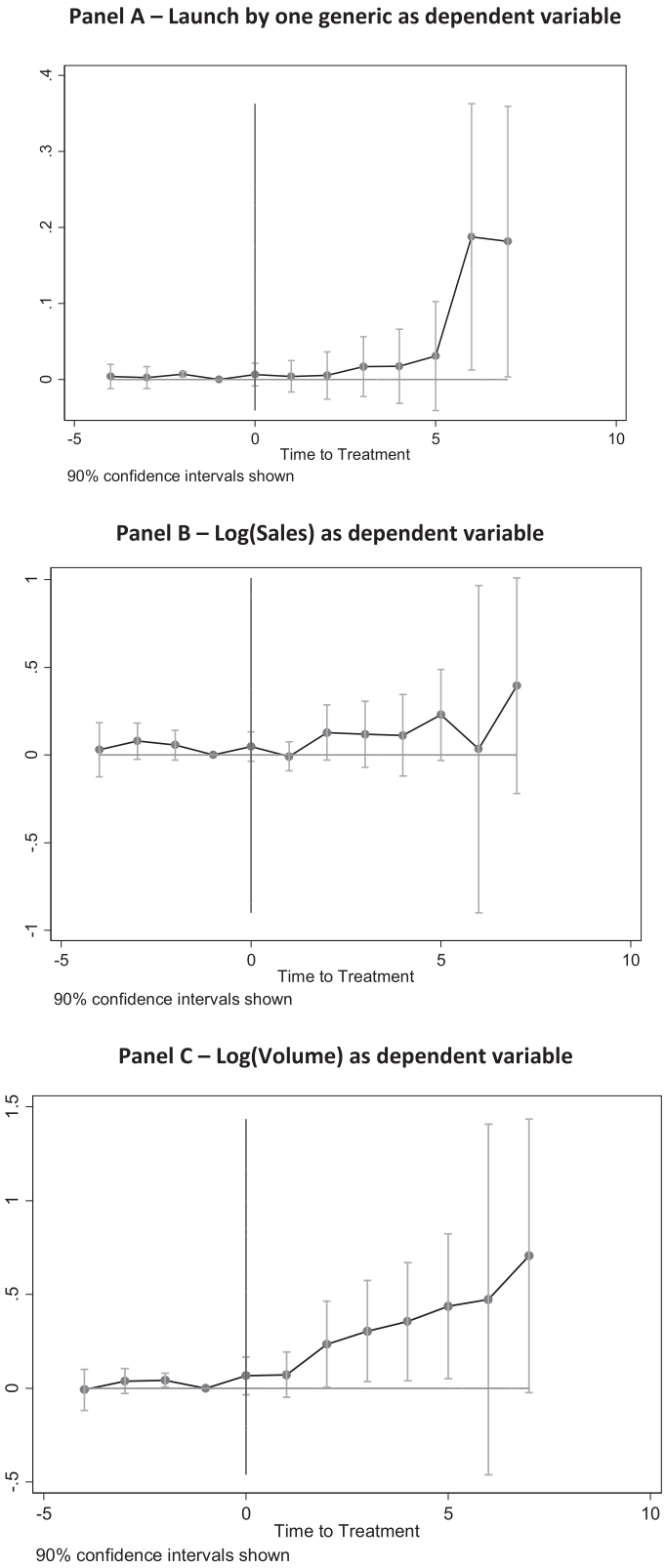
[11]To gain the intuition for this general feature of geographic pools, consider the following illustrative example. A drug is available for license in two countries: Albania and South Africa. Assume that in the absence of MPP there are no licensees in Albania and four licensees in South Africa. Assume that both countries are included in the MPP license for the drug and that three sublicensees sign up. The number of deals increases by three in both countries, but the dummy Access changes from zero to one in Albania and is not affected in South Africa.

[12]This is an important difference with recent studies examining the effects of the MPP (Martinelli et al., 2022; Wang, 2022). They focus on transactions that rely on resources from international organizations, such as the Global Fund, which predominantly operate in Sub-Saharan Africa. The IQVIA data include local sales financed by these organizations as well as other local sales to local governments and private entities.

[13]Because many Sub-Saharan and poor countries that are often included in MPP agreements are not present in the IQVIA data, the dummy MPP is equal to one for only 3.6% of the sample (compared to 7% in the licensing sample). For about 75% of the sample, we observe no firm active in the product-country, that is, no firm has launched the product in the country by that year. On average, at least one licensee launches a product in about 7% of the observations. Despite the smaller sample, the relationship between licensing and inclusion in the MPP replicates well in the IQVIA sample. The coefficient in the regression of Access on the MPP dummy is 0.691 (p-value < 0.01), and for the number of deals the coefficient 4.241 (p-value < 0.01). These are similar to the baseline estimates in column 1 and 2 in table 2.

[14]Volume is constructed at the product level adjusting the IQVIA data on packs sold for the number of vials or tablets in each pack.

FIGURE 2.—MPP AND MARKET OUTCOMES



This figure plots the estimates of a regression in which the effect of the MPP is separately estimated for each year before and after inclusion. The figures plot the coefficients (and 90% confidence intervals) with the year before inclusion normalized to zero. Standard errors are clustered at the product and country level. The sample is the full IQVIA data set. The dependent variable Launch by one generic is equal to 1 if at least one licensee has commercialized the drug in the country in year *t*. Sales are in U.S. dollars, and volume is in tablets.

rising by about 60% by the fifth year. Some of this increase in the early years may be due to volume associated with firms that entered before the product was included in the MPP. We cannot distinguish the source of the increase in volume.[15] Together, the null effect on sales and the positive effect on volume suggest that the underlying demand for life saving drugs has roughly a unit elasticity. This is consistent with the estimates for HIV drugs in Kremer and Snyder (2015, 2018), and Dubois et al. (2021), who study the impact of drug procurement systems on prices in low- and middle-income countries.[16]

In summary, the evidence indicates that entry into the patent pool promotes product launches and adoption of essential medicines. The limitations of the IQVIA data do not allow us to pin down the effects with precision. Nonetheless, the results point to a potential gain in consumer welfare from the MPP, though the required product launches appear to take several years to materialize.

B. *Discussion and Potential Welfare Effects*

We have shown that entry into the MPP sharply increases the likelihood of observing at least one licensee for the product-country. Our simple model explains this as being driven by a reduction in transaction costs. An alternative explanation, however, is that the patent pool may offer more favorable licensing terms than bilateral licenses. We do not have information on the terms and conditions of the bilateral deals in our sample to test this hypothesis directly. However, we expect the differences in licensing terms not to be substantial for several reasons. First, media statements by pharmaceutical companies after the implementation of bilateral agreements indicate that royalties are often low or zero.[17] Second, many of the bilateral deals in our sample focus on Sub-Saharan Africa or other low-income countries, which have limited potential to generate royalty income in any case. Finally, some anecdotal evidence suggests that the MPP designed some of its downstream licenses using drug companies' bilateral contracts as templates.[18] In view of these considerations, our findings about the impact of the MPP appear more consistent with the transaction cost interpretation.

[15]With the available data, we cannot identify the effect of the MPP on the number of entrants for a given product-country. For about 70% of product-country-years in which entry occurs, there is only one active firm. A more thorough analysis of this issue will require richer data on market outcomes.

[16]The range of demand elasticity estimates in Dubois et al. (2021) vary between −0.73 and −0.97. Their analysis uses a wider set of molecules than the one in our data (including, but not limited to, HIV and tuberculosis) but only seven countries.

[17]For example, the 2011 license between Bristol Myers Squibb and Matrix Laboratories for stavudine and didanosine in Sub-Saharan Africa is royalty free. The same is true for a number of Boehringer-Ingelheim's licenses of nevirapine in low-income countries.

[18]This led some advocacy groups, such as the Initiative for Medicines, Access, and Knowledge (I-MAK), to voice concerns after the 2011 MPP-Gilead agreement because it appeared too similar to existing Gilead licenses. Source: www.i-mak.org/2011/10/11/implications-of-the-patent-pool-licenses-with-gilead-part-ii/.

Our analysis also shows that the likelihood of a product launch in the country increases after entry into the MPP. However, the size of this effect on launch is much smaller, and more delayed, than the impact on licensing, which is consistent with the theoretical discussion in section IV. To get a better sense of the relative size of the effects on licensing and launch, our baseline regression (column 1, table 2)—evaluated at the means of covariates and the estimated fixed effects—implies that the probability of observing at least one license is 0.11 for product-countries that are not in the MPP, and 0.84 for those in the MPP. Using estimates from an unreported regression of launches on *Access,* we find that having one licensing deal translates into an actual launch with probability 0.62 for product-countries that are not in the pool and 0.12 for those covered by the MPP. Together, these estimates imply that, in country-products not covered by the pool, the probability of observing one launch is about 0.07 ($= 0.11 \times 0.62$), but it is 0.10 ($= 0.84 \times 0.12$) for product countries included in the MPP. In short, inclusion in the MPP increases the probability of licensing by 73 percentage points (from 0.11 to 0.84), but it increases the probability of observing an actual launch by only 3 percentage points (from 0.07 to 0.10). This finding has an important implication for empirical research on patent pools with bundled licensing, as it shows that focusing on licensing can be very misleading as to the impact of the pool on actual entry and diffusion.

An important caveat is that more than half of the MPP product-countries in our sample were included in the pool in the last five sample years. Product launches take several years, as commercialization requires setting up manufacturing and distribution facilities as well as obtaining WHO prequalification of the generic version of the drug. Thus, censoring may lead us to underestimate the impact of the MPP on launches in the baseline specification where we use a linear probability model. To address this issue, in appendix A5 we analyze the link between MPP inclusion and product launches using hazard models. As predicted, we find larger effects in these specifications which allow for censoring. This provides further support for our conclusion that the MPP increases the likelihood of actual launches, not just licensing.

Finally, while bearing in mind the serious data limitations for our analysis of market outcomes, we provide a "back of the envelope" calculation of the welfare gains from the MPP. As explained earlier, our findings on sales and volume imply an underlying (roughly) unit elastic demand. This takes the form $p = A/q$, where A is the total sales revenue obtained in the market. When at least one unit of product is consumed, the corresponding total welfare is $W = A\log(q)$, which implies that an increase in quantity by z percent translates to a welfare gain of $Az/100$. To make this computation, we need to know the sales generated by a drug after it enters the pool and use our parameter estimates of the MPP impact on volume to compute the welfare generated by each drug licensed by the MPP. This computation is likely to be a substantial underestimate, because many of the countries in the MPP are not in the IQVIA sales data, and many of the MPP licenses are in our data for a very short time span. Doing this computation on the available data, we estimate a welfare gain of roughly 27 million USD for the period 2010–2018. By comparison, the total operating costs of the MPP for the period 2010–2015 was 22.9 million USD (Juneja et al., 2017). Even this lower-bound estimate suggests that the welfare gains from the MPP may exceed the cost of the institution. However, a more complete welfare analysis requires richer data in terms of launched products and a longer time series on market outcomes.[19]

IX. Concluding Remarks

In this paper we examine how the Medicines Patent Pool affects licensing and launch of HIV and other essential drugs in low- and middle-income countries. We have three key empirical findings. First, inclusion in the pool is associated with a large increase in the probability of licensing. Second, this effect is heterogeneous: it is smaller in middle-income countries with large exposure to HIV (where bilateral deals are more likely). Finally, some evidence indicates that the MPP increases not only licensing, but also the likelihood of launch and the quantity of these drugs sold in the market. However, the effects on launch and market outcomes are much smaller than on licensing, as a consequence of the geographical bundling of license contracts by the patent pool. This finding highlights that, to study the impact of such patent pools on technology diffusion, it is important to go beyond their impact on licensing, which, by itself, is likely to overstate their impact. More broadly, our paper suggests the potential of pools to promote the diffusion of essential vaccines and other types of innovation in developing countries.

[19]To perform this computation we identified the product-countries in which MPP licenses were in place for more than five years and for which positive sales were recorded by IQVIA. We used the average revenue generated in the sample years following the fourth after MPP inclusion as a proxy for the demand scale parameter, A. The increase in quantity is obtained by exponentiating the coefficient estimated for the fifth year after MPP inclusion in the regressions underlying panel C in figure 2 (61%), and we assume this increase in volume lasts until the patent expires in the focal country.

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